

Problem Set 10: Model Comparison

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Test Set Performance

Model	Accuracy
Logistic Regression (LASSO)	0.853
Decision Tree	0.868
Neural Network	0.858
K-Nearest Neighbors	0.843
Support Vector Machine (RBF)	0.860

Table 1: Out-of-sample accuracy across classification models

Best Performing Model

The decision tree model achieved the highest test accuracy (0.868), making it the best-performing model in this comparison. The K-Nearest Neighbors model had the lowest test accuracy (0.843).

The Support Vector Machine (0.860) and Neural Network (0.858) performed very similarly. The Logistic Regression (LASSO) model achieved an accuracy of 0.853, which is slightly lower but still competitive compared to the other models.

Overall, the performance differences across models are relatively small, suggesting that multiple classification methods perform reasonably well on this dataset.